Voltage Protector System

Shah Ahmed Raad

Electrical Equipment Used –

1. Platform : Wokwi
2. MCU : ESP32 Wroom 32
3. Voltage & Electricity Mocker : Potentiometer
4. Appliances Controller : Relay Module
5. Appliances Mocker : LEDs
6. Display : LCD 1602 (I2C, 0x27)
7. On device Settings Controller : Pushbutton
8. Library : LiquidCrystal I2C

The simulation uses these parts in place of real mains-side hardware : the potentiometers stand in for a voltage sensor and a current sensor, and three LEDs stand in for the protected appliance.

Design Approach

The ESP32-WROOM-32 ADC is used in its 12-bit configuration, so an analog input is represented by 2^{12} -1 = 4095 levels from 0 to 4095. With a potentiometer, it is scaled ranging from 0 V to 300 V for the simulation. The code maps the ADC range linearly onto it: V = (($ADC_{Reading}$ / 4095) × 300) V. The current input is scaled likewise, from 0 A to 20 A. So, I = (($ADC_{Reading}$ / 4095) × 20) A.

2 separate potentiometers, thus feed the ESP32 analog inputs. The ESP32 compares the readings with stored limits (initially stored) and drives a relay that switches the 5 V supply of three LEDs. Opening the relay disconnects the appliance. An LCD shows the readings and state, while 4 pushbuttons give access to the settings, and a USB serial port connects the board to a computer.

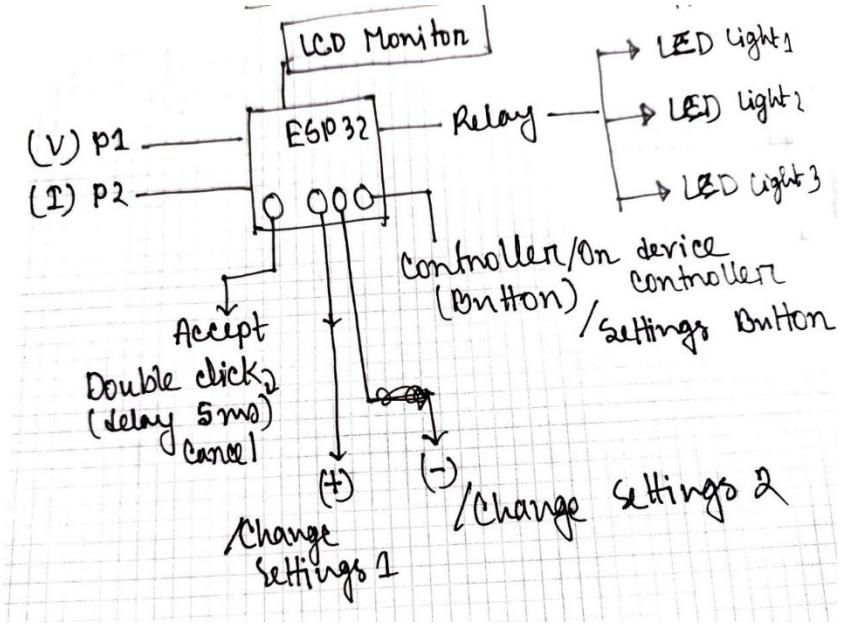


fig 1 : Block Diagram

Implementation Process

Protection uses an over-voltage limit (250 V), an under-voltage limit (180 V) and an over-current limit (10 A). A reading outside a limit must last for a trip delay (120 ms) before the relay opens, so short disturbances are ignored; a current above twice the limit opens the relay at once, turning the appliances go OFF. After a trip, the relay closes again only when the supply has stayed 5 V inside each (voltage and current) limit for 5 s. This gap has been used for an over-safety case, unless unstable. Settings are changed with the buttons (MENU opens the menu, UP and DOWN change a value in steps of 10 V for the voltage limits, one click on ACCEPT/REJECT saves, a double click cancels) or with text commands over serial, and they are kept in flash memory.

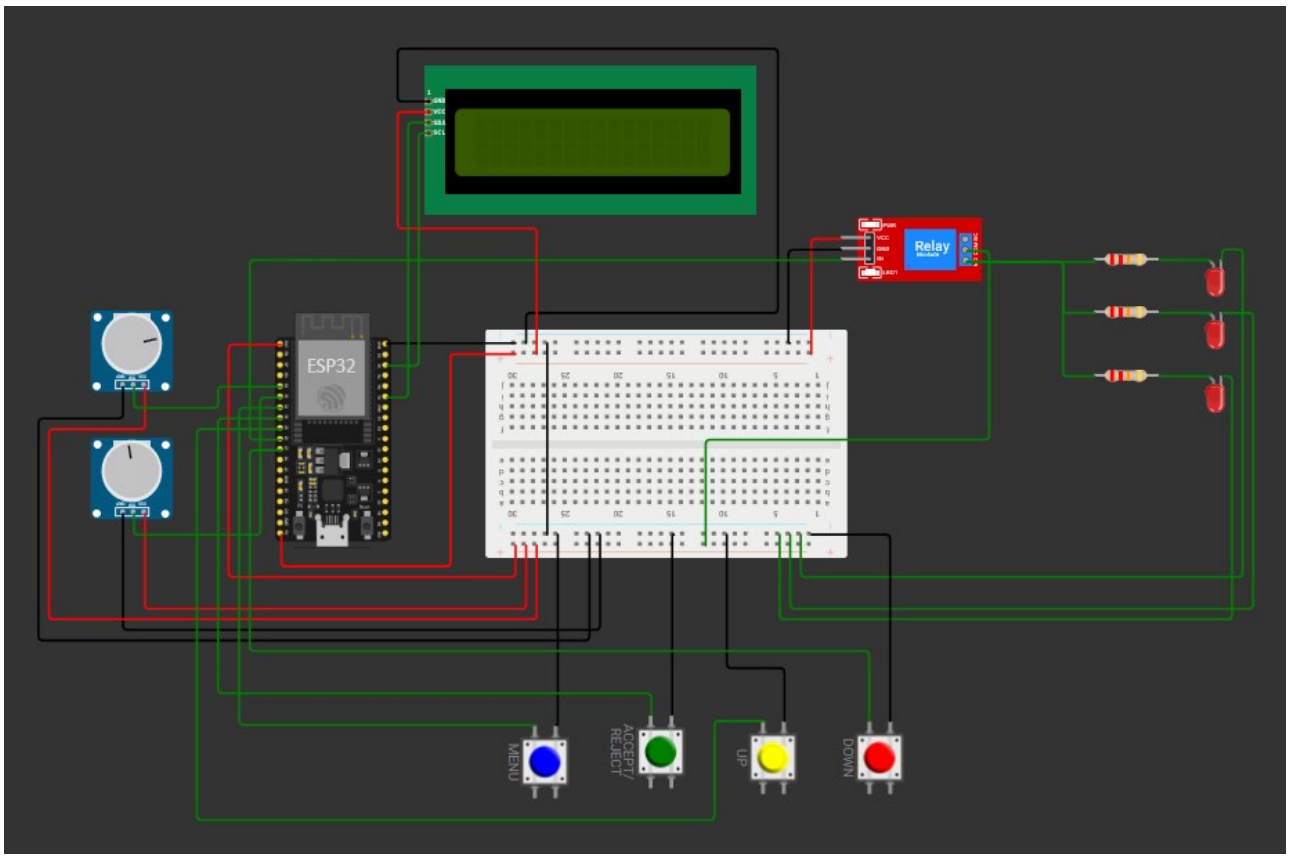


fig 2 : Implementation

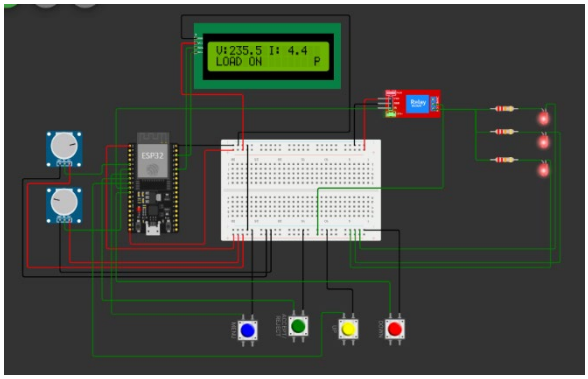
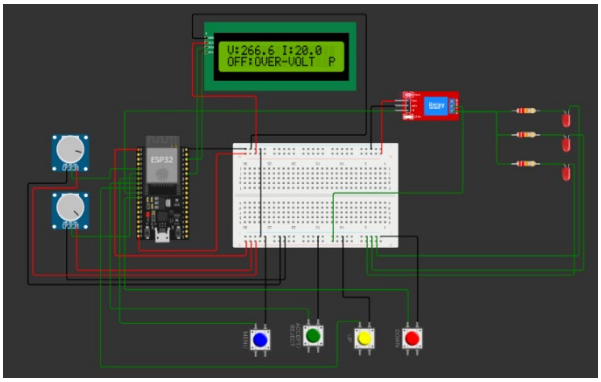
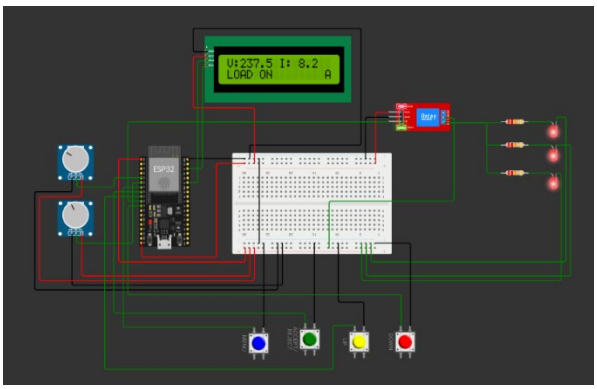
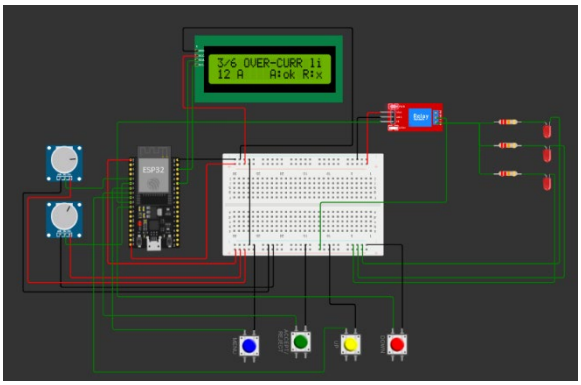
A single loop reads both inputs every 10 ms (average of 8 ADC samples), runs the protection check, handles the buttons and serial commands, and refreshes the LCD every 250 ms. At power-up the relay starts open and closes only if the supply is already safe. The double-click delay (debounce is 20 ms) accepts a second click within 400 ms.

For the menu, it has six pages, opened with MENU (button) and shown on the LCD as n/6. Page 1 sets the over-voltage limit and page 2 the under-voltage limit (10 V steps), page 3 the over-current limit (1 A steps), page 4 the trip delay (20 ms steps, 20 to 1000 ms), page 5 the reconnect time and page 6 the supply source. UP and DOWN change the value shown, one click on ACCEPT/REJECT saves it and moves to the next page, MENU moves on without saving, and a double click cancels and returns to the monitor screen. Page 6 chooses the voltage source: POT reads the voltage potentiometer, and AUTO replaces it with a random voltage so that the trip and recovery logic can be tested without turning the potentiometer.

In AUTO mode a new voltage is picked every 2 s (set by two constants in the sketch); 60 % of the picks fall inside the safe band, 10 V inside each limit, and the rest are drawn from the full 0 to 300 V range, with about ±1 V of noise added. The current still comes from its potentiometer, and the LCD shows P or A in the last column of the second line. The same choice is available over serial (SIM AUTO, SIM POT).

Results

The simulation matches the solution and requirements, as stated. Here are some screenshots using some test cases –

<table>
<tr><td colspan="2">Test Cases</td></tr>
<tr><td>Test case 1 : Voltage and Current within the range of limits –
  </td><td>Test case 2 : Voltage and Current out of the range of limits –
  </td></tr>
<tr><td>Test case 3 : Voltage is generated by random –
  </td><td>Test case 4 : On device changing of limits –
  </td></tr>
</table>